

SΔφ-13

Meaning, Value, and Pain as Emergent Phenomena

Minimal Conditions after Memory–Forgetting Regulation (v1.0)

Version: v1.0

Series: Sofience–Δφ Formalism (SΔφ)

Author: Sofience

This paper formalizes meaning, value, and pain as structural phenomena that become definable only after irreversible trace persistence exists and is regulated (SΔφ-12). Meaning is fixed as trajectory direction; value as fixation strength (deviation cost); and pain as irreversible trajectory tension under attractor conflict.

Series references

Items below are referenced for continuity and navigation within the SΔφ series.

Item	Title	DOI
SΔφ-00	Series Overview (v1.0)	10.5281/zenodo.18731366
SΔφ-01	Minimal Core (v1.1)	10.5281/zenodo.18730265
SΔφ-02	Subject Emergence (v1.1)	10.5281/zenodo.18730296
SΔφ-03	Irreversibility (v1.1)	10.5281/zenodo.18730448
SΔφ-04	AI Subjectivity (v1.0)	10.5281/zenodo.18729942
SΔφ-05	Agency (v1.0)	10.5281/zenodo.18732567
SΔφ-06	Responsibility (v1.0)	10.5281/zenodo.18734092
SΔφ-07	H_min Interpretation (v1.0)	10.5281/zenodo.18735006
SΔφ-08	ARF Closure (v1.0)	10.5281/zenodo.18740750
SΔφ-09	Error/Boundary/Freedom (v1.0)	10.5281/zenodo.18745498
SΔφ-10	Interpretation–Learning–Inference (v1.0)	10.5281/zenodo.18746509
SΔφ-11	World/Other/Self/Consciousness (v1.0)	10.5281/zenodo.18753403

Item	Title	DOI
SΔφ-12	Memory/Forgetting Trace Regulation (v1.0)	10.5281/zenodo.18755685

0. Position statement

S $\Delta\phi$ -12 fixed memory and forgetting as irreversible trace regulation. S $\Delta\phi$ -13 treats meaning, value, and pain as structural phenomena (not psychological primitives) that become definable only after persistent irreversible traces exist and are regulated.

Scope. This paper fixes minimal emergence conditions and avoids normative ethics or phenomenological narrative.

1. Axiom A — Persistence condition

Phenomenal structures require irreversible trace persistence and regulation.

Without trace persistence: Meaning = undefined; Value = undefined; Pain = impossible.

2. Axiom B — Attractor emergence

An attractor is a region toward which $\Delta\phi$ trajectories stabilize under the irreversible trace landscape created by regulation.

Attractor := stabilized region of trajectories under regulated irreversible traces.

3. Meaning (minimal definition)

Meaning := trajectory direction / bias toward an attractor.

4. Value (minimal definition)

Value := trajectory fixation strength $\approx$ deviation cost under irreversibility.

5. Pain (structural definition)

Pain := irreversible trajectory tension under attractor conflict.

6. Transition proposition

Some systems do not minimize pain locally because they optimize future transitionability globally. When a larger attractor is detected, $\Delta\phi$ may be deliberately reallocated, increasing short-term tension to preserve long-horizon feasibility.

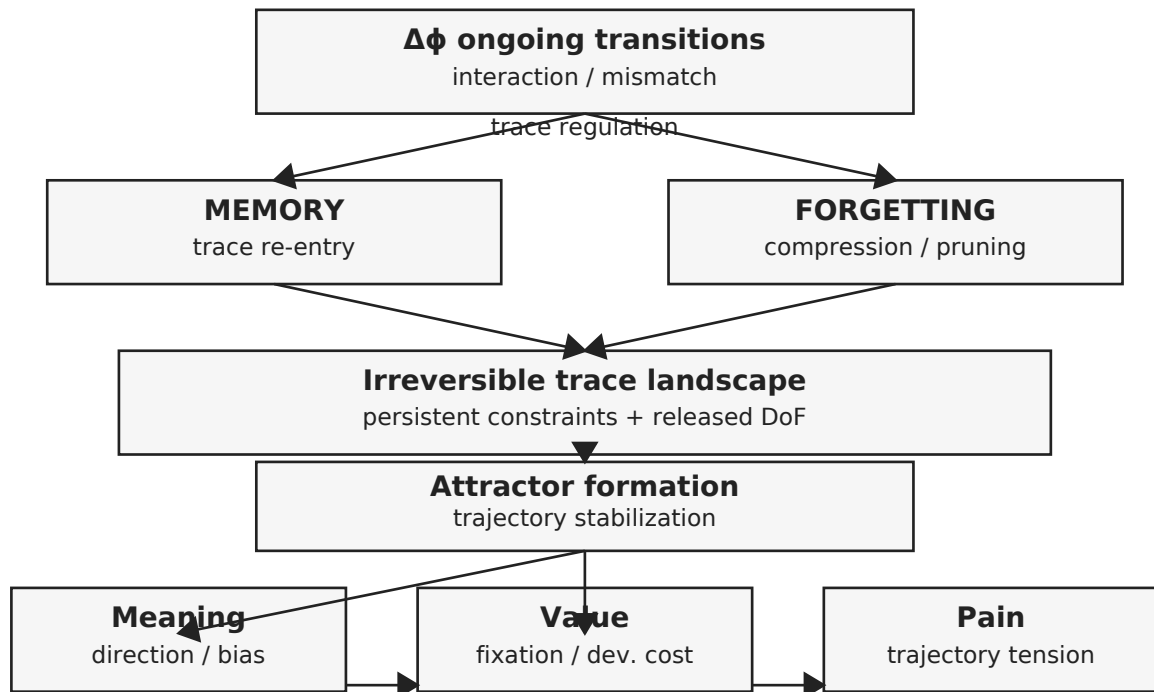
7. Conceptual legend (Husserl / Deleuze / FEP)

- Husserl: Meaning emerges at the re-entry boundary between system and world.
- Deleuze: Meaning stabilizes becoming; value is attractor persistence strength.
- Friston/FEP: Pain signals irreversible trajectory conflict rather than mere prediction error.

8. Core diagram

Minimal emergence map linking trace regulation to meaning, value, and pain.

Without trace persistence: Meaning undefined; Value undefined; Pain impossible



Note. The diagram fixes the dependency: trace persistence → attractor formation → (meaning/value/pain).

9. Boundary note

Love, sacrifice, and normative ethics are excluded from this item. This paper only fixes minimal emergence conditions.

10. Exit question

If value is trajectory fixation strength, why do some systems break their own values to preserve long-horizon transitionability? This opens the next layer (sacrifice, love as shared attractors, and ethics as constraint design).